

Yuriy Korotovskikh

SENIOR FULL-STACK ENGINEER (FRONTEND-FOCUSED) · WEB3 · FINTECH

Yerevan, Republic of Armenia

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Summary

Senior Frontend / Fullstack Web3 Engineer building wallets, payment systems and dApp flows since 2020.

I work where product UX meets blockchain complexity: wallet connection, approvals and signing, and precision-sensitive product logic.

Built payment flows, trading systems, dApps, block explorers, indexers, developer tools, SDKs, and reusable production-grade UI systems across Web3 and fintech.

Shipped products such as browser smart contracts playground, block explorers, CLIs, several production dApps, taking them from early-stage ideas to reliable releases with a focus on usability and delivery.

Skills

Languages Proficiency - JavaScript, TypeScript, Solidity, HTML, CSS, **Working Knowledge** - Go, Python

Frontend

React, Next.js, Astro, Vue.js, Redux, Effector, Zustand, Web Workers, Storybook, Vite, Rollup, Tailwind, WebAssembly, TanStack, Frontend architecture

Web3

Viem, Wagmi, Ethers, MetaMask, WalletConnect, Ledger SDK, RainbowKit, Reown AppKit

DevOps

Docker, Ansible, TeamCity, Nix, GitHub Actions CI

Backend

Node.js, SQL, REST, JSON-RPC, WebSocket

Testing

Jest, Vitest, Cypress, Playwright, Synpress, Hardhat, Stryker

UX

Interaction design, Figma, Design systems, User research, Accessibility

Work Experience

Mercuryo

London, United Kingdom (remote)

SENIOR FRONTEND DEVELOPER

Sept. 2025 - Current

Mercuryo is a crypto-powered payments infrastructure provider whose core products include a wallet-integrated on/off-ramp widget, Spend Card, and Wallet (10M users as of May 2025).

- Owned the Mastercard Crypto Credentials frontend, including wallet linking, message and transaction signing, and error-handling flows across partner wallets.
- Architected and owned the Boost Points Rewards frontend flows, including the UI, advanced Redux state management, event analytics, multi-partner compatibility, and backend integration.
- Led frontend observability, incident response protocols, security and code quality initiatives, including Sentry instrumentation, product analytics, Snyk monitoring, dependency management, static analysis tooling (zizmor, React Doctor, ESLint, Oxc), and nightly CI checks.
- Led frontend onboarding and learning initiatives through internal workshops and documentation, and regular updates on frontend trends and best practices.
- Delivered end-to-end frontend flows for Spend Card: eligibility checks, multi-level KYC flows, Sumsub integration, card issuance, top-ups, limits, transaction history, balance states.
- Integrated Apple Pay, Google Pay, and Cloudflare Turnstile into Mercuryo frontend flows.
- Built and maintained partner wallet integrations (Ledger, MetaMask, Trust Wallet, and others) in a React + Redux widget with white-label embedding, theming, advanced customization, localization, and iframe-host communication.
- Integrated core Web3 UX primitives: WalletConnect/Reown AppKit connection, multi-chain EVM support with viem and wagmi.
- Expanded automated test coverage for high-risk payment flows using Playwright E2E and Vitest unit tests, and built a sharded Playwright pipeline on self-hosted runners with Allure reporting.
- Shaped frontend architecture with FSD-inspired module boundaries, separation of concerns, dependency flow rules, and code quality gates to keep a large frontend codebase maintainable. Contributed to frontend engineering standards, and scalable development practices across the team. Actively participated in architecture migrations and refactoring.
- Built AI-assisted engineering workflows by authoring multi-level agent guides, harness tooling, and several reusable internal skills for code generation, test authoring, and review checklists.

Diffuse Prime

Limassol, Cyprus (remote, contract)

FOUNDING FRONTEND ENGINEER

Aug. 2025 - Mar. 2026

Diffuse Prime is a ZK/TEE-powered non-custodial lending protocol for structured leverage deals.

- Designed architecture and built a Next.js + TypeScript frontend dApp from scratch, including wallet connection, borrow/lend flows, tx simulation, approvals, stale on-chain state handling, and position state for lenders and borrowers. Integrated Sentry, logging, and event analytics to monitor the product.
- In collaboration with the design team, implemented a modern ESM UI kit with Radix UI, Tailwind, and TanStack libraries.

- Implemented a testing strategy with Vitest unit tests, visual regression tests, Lighthouse tests, and Playwright end-to-end tests with local network fixtures. Covered more than 90% of the codebase with tests.
- Designed and implemented full cycle CI/CD pipelines for the frontend, ensuring integration and delivery to Vercel.
- Designed and implemented a Node.js indexer for processing on-chain events, maintaining position state, and keeping event handling idempotent, then integrated it with the Next.js API and Vercel cron jobs.
- Implemented a custom JavaScript SDK for interacting with the protocol, encapsulating precision-sensitive protocol calculations (WAD math and integer arithmetic) for the frontend and other tools.
- Organized monorepo with npm workspaces, implemented automated release and versioning process with changesets, introduced best security practices and audits, and set up GitHub Actions CI.
- Collaborated closely with founders as a sole frontend engineer to design and implement various features and visual elements, to ensure the product meets the needs of its users.
- Designed repository AI workflows by authoring agent operating rules, building task-specific agents (feature delivery, refactoring), and creating reusable prompt patterns for consistent output quality.

Nil foundation

Limassol, Cyprus (remote)

FRONTEND DEVELOPER

May. 2022 - Aug. 2025

Nil Foundation is an Ethereum L2 startup working on Ethereum horizontal scalability.

- Built from scratch and released proof market DEX-like interface, which is a decentralized exchange of computation powers for proofs generation.
- Built a block explorer for the NIL Layer 2 chain with detailed, data-heavy views for blocks, transactions, and accounts. Designed and optimized the frontend architecture for performance and scalability, and contributed to the Node.js backend powering the explorer.
- Developed an interactive, sharding-specific smart contract playground for the NIL chain. The tool enables users to write and compile Solidity code, save snippets, deploy contracts, and interact with them — all within a unified interface.
- Built a custom JavaScript SDK for interacting with the Nil blockchain, tailored to support SSZ serialization and Poseidon hashing. This library became the foundation for multiple critical tools, including a CLI, Hardhat plugin, and browser wallet extension.
- Developed and maintained the corporate website using Next.js with a Strapi CMS backend, along with additional marketing pages to support product outreach.
- Created and maintained a reusable React UI component library, adopted across nearly all NIL interfaces to ensure design consistency and development efficiency.
- Built and maintained CI/CD pipelines using GitHub Actions CI, Ansible, and Jinja.

InfoTeCS JSC

Saint Petersburg, Russia (on-site)

FRONTEND DEVELOPER

Mar. 2021 - May. 2022

InfoTeCS is a cybersecurity company providing a range of secure communication solutions for corporate clients.

- Built Remote Work Management System, a big internal React app, where employees could request remote work, track approval pipelines, and submit daily reports. I created interfaces, pages and forms, worked on performance optimizations and managed API workflows with Redux Saga. I also contributed to backend C# controllers and validations to support frontend logic.
- Contributed to Employee performance evaluation system - a large-scale web application built with ASP.NET and a legacy frontend stack using jQuery and Backbone.js.
- Developed React-based Artifact Management App. This internal tool allowed teams to upload, download, and manage build artifacts across multiple systems. All existing artifacts were aggregated in a dynamic responsive table with server-side sorting and filtering, making it easy for engineers to quickly locate, and interact with the files.
- Led frontend development for a Next.js documentation portal targeting foreign branches, combining static site generation with dynamic internal routing. This portal was a key touchpoint between the company and its foreign customers. One part of the app was public-facing, the other integrated into an internal doc system.
- Maintained React components library and closely collaborated with Design department to ensure consistent implementation of the design system.

Oday Technologies

Moscow, Russia (remote)

FRONTEND DEVELOPER

Sept. 2020 - Mar. 2021

Oday Technologies is a company specializing in the development of automation tools and information security.

- Designed and developed a Vue 2 and Vuetify-based monitoring service, integrating interactive Leaflet maps for real-time data visualization and location tracking.
- Worked on a large financial analytics React application with a Python backend, where I created new UI elements, optimized loading time, and implemented advanced search across multiple sources.

Education

Kurgan State University (KSU)

Kurgan, Russia

BACHELOR'S DEGREE

Sep. 2014 - Aug. 2018

Saint Petersburg State University of Economics (UNECON)

Saint Petersburg, Russia

MASTER'S DEGREE

Sep. 2018 - Jul. 2020